

Peer review report

Manuscript: “An IoT Enabled Smart Hygiene Monitoring System for Enhanced Railway Restroom Sanitation Management”

Journal: *Discover Sustainability*

Recommendation: Major revision

I have read this revised version carefully, and I want to begin with what the work does well. The problem it tackles —hygiene in railway restrooms— is real, clearly framed, and a good fit for the journal. The prototype exists, it works, and the authors explain its operating logic in welcome detail; the eight operating scenarios are easy to follow. That said, my overall impression is that the manuscript is not yet ready for publication. Three of its foundations —the empirical motivation, the validation of results, and the sustainability argument itself— need substantial strengthening, and for that reason my recommendation is major revision.

On the response letter, the two points raised are resolved in form: the title now reads as a single sentence without punctuation, and the data availability statement has been added. Let me add one remark on the latter. Since the results are essentially screenshots and a latency table, the wording chosen —“*The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.*”— is the bare minimum acceptable. I would encourage the authors to deposit the raw measurement data in a repository with a persistent identifier; for a journal centred on sustainability and openness, that adds credibility.

My main concern is the empirical basis of the paper. The motivation rests on a survey the text introduces as “*A survey was conducted regarding the cleanliness and hygiene in railways with the support of 10K+ respondents*”, and from which specific figures are quoted —“*68 percent of the respondents have the major issue with dirty restrooms, whereas 46 percent of them were unhappy about rare or no cleaning of the restrooms*”. The trouble is the source: reference [28] is “*Smart Hygiene Monitoring System Demo: Project Demonstration Resource*”, a Google link (share.google/SC3YFAFA7vn7RhBTo), and the bar chart itself appears to be generated from it. None of this is traceable or verifiable, and building the problem statement on it is, to my mind, untenable. I would ask the authors to replace that reference with a real, citable source —railway administration reports, published studies— or, failing that, to remove the percentages and reframe the motivation in qualitative terms.

The second issue, a scientific one, is validation. The results section opens by claiming it presents “*validated outcomes produced from the multiple runs of the developed model under varied conditions*”, but I find no figures backing the word “validated,” and the abstract offers no metric either. There is no statement of how many repetitions were run, no accuracy of the usage count, and no false-positive or false-negative rates for occupancy detection. That last gap is striking because the text itself concedes the sensor’s fallibility —“*It can give a false detection due to reflections, angled surfaces, soft absorbent targets, or nearby non-person objects inside the threshold*”— without reporting how often it happens. The latency table compounds the impression: it reports values to the hundredth of a second (0.42 s; 0.58 s; 1.42 s...) without explaining how they were measured, with what instrument, or over how many trials. Reporting a mean and standard deviation over a stated number of runs would be enough to give “validated” some substance.

Because this journal weighs the contribution to sustainability above all, I dwell on that as well. The framing —SDG-6, WASH, water conservation— is asserted but never quantified. I miss an estimate, even an order-of-magnitude one, of water per cleaning cycle, of the saving relative to scheduled cleaning, or of energy use; and the repeated claim that the system is low-cost (“*a low-cost ecosystem*”) should be backed by at least a bill of materials with prices. As it stands,

the sustainability contribution is more declared than demonstrated, and that is precisely the axis on which the work is judged.

A related point: the novelty claim should be calibrated. The authors’ own comparison tables show prior systems with more capabilities (environmental monitoring, AI analytics) than the one proposed here, which is essentially a subset: door, occupancy, and a manual pump. I suggest reframing the contribution candidly —its value lies in low cost and a “human-in-the-loop” approach, not in functionally surpassing the state of the art.

One point gave me pause and deserves to be addressed head-on. In the discussion the authors candidly admit that staff can silence the buzzer without actually cleaning —“*may also cause the cleaning staff to just turning off the buzzer and not performing the manual cleaning process*”—, that the system does not supervise cleaning quality, and that, running on a local network, a component failure goes unreported to the supervisor. I value that transparency, but those same limitations erode the central claim that the system improves maintenance reliability. I would ask the authors to reconcile the tension: either temper the scope of the contribution, or propose a mechanism —a log of cleaning actually performed, an operator confirmation— that closes the gap.

There is also a set of internal inconsistencies worth cleaning up, since they give the impression of a text not fully revised. I flag each with the exact fragment from the document:

- Two tables are numbered “Table 2” (the comparison table on p. 5 and the latency table), and inside the paragraph describing the sensors the line “*Table 2. A detailed latency description table for various modules and its operations*” is left loose; it repeats the Table 3 title and is mis-numbered.
- Reference [22] appears twice in the comparison table with contradictory descriptions, once as “*Train toilet sanitation*” and once as “*UV sanitization*”; and the row for [20], which is actually a paper on thermal-sensor light control, is labelled “*Campus hygiene system*”.
- In Scenario 6 the text ends with “*indicating ‘Door Closed, No Person Inside’ as shown in Fig. 10*” when the relevant figure is 11.
- In the HC-SR04 description a stray sentence with no subject or verb is left behind: “*±3 mm within the specified operating range under proper conditions.*”
- Naming wavers: the microcontroller is “*ESP WROOM-32*” in the text and “*ESP32 Controller*” in the block diagram, and the door sensor goes from “*Hall-effect sensor*” to “*SS49E Linear Hall Sensor*”. A legend explaining the single and double check marks in the tables is also missing.

On the bibliography, I found a couple of inaccuracies to correct. Reference [7] is cited as “*Arduino-based automated washroom sanitizing system. International Journal Publication (2023)*”, but the DOI resolves to a **2020** IEEE ICICCS conference paper, and its fourth author is not “Francis, A.”. Reference [1] opens with “*Jabbi, B.A.*” when the first author is Alhaji Jabbi (initial “A.”), and one co-author is omitted. These are minor, but I take the chance to recommend that the authors check the full reference list with the same care.

Finally, the manuscript needs thorough, professional language editing. There are many ungrammatical or confusing constructions; a few examples taken verbatim from the text: “*There is a rapid increasing in growth of population in urban localities*”; “*the usage count isn’t until the person fully enters the premises*”; “*even id the push button is pressed externally*”; “*further incense of hygienic conditions*”. These stumbles hinder reading and a sound piece of technical work should not have to carry them.

I will close by stressing that the underlying idea is valuable and, in my view, salvageable. If the authors ground the motivation in a verifiable source, provide quantitative validation data, quantify —if only approximately— the sustainability benefit, and clean up the text, the manuscript can improve markedly. I look forward to a revised version.